Relating SOA and Cloud Computing

- Cloud computing is still in its infancy, and although Web services can implement a Service Oriented Architecture, it is not a requirement.

- Applications of those types have less of a need for the flexibility and loose coupling that SOA provides.

- As cloud applications become more diverse in scope, SOA offers an architectural blueprint for accessing diverse optimized services through a loosely coupled standardized method that provides an ability to evolve that is difficult to implement in any other way.

- SOA is loosely coupled because the service is separated from the messaging. If a component doesn’t provide the capabilities required, it is an easy task to switch to a different component, and switching requires almost no programming.

- Developers lose some of the ability to customize modules, but gain a significant advantage in simplifying their applications. Taken as a whole, applications that rely on SOA components can be very complex and appear to be tightly coupled, when in reality they are not.

SOA Security

- Any system that sends hundreds or thousands of messages across an internetwork as SOA does is subject to attack in all the traditional ways that network traffic is hijacked, spoofed, redirected, or blocked.

- Because SOA eliminates the use of application boundaries, the traditional methods where security is at the application level aren't likely to be effective.

- Cisco has a family of products that enforce rules and policies for the transmission of XML messaging that they have named Application Oriented Networking (AON; http://www.cisco.com/en/US/products/ps6480/).

- A similar policy based XML security service may be found in Citrix's NetScaler 9.0 (http://www.citrix.com/English/ps2/products/product.asp?contentID=21679) Web application delivery appliance.

- To address SOA security, a set of OASIS standards (http://www.oasis-open.org/committees/tc_home.php?wg_abbrev=security) was created, which includes the following:

SOA Security

- **Security Assertion Markup Language (SAML)** is an XML standard that provides for data authentication and authorization between client and service.

- The SAML technology is used as part of Single Sign-on Systems (SSO) and allows a user logging into a system from a Web browser to have access to distributed SOA resources.

- **WS-Security (WSS)** is an extension of SOA that enforces security by applying tokens such as Kerberos, SAML, or X.509 to messages.Through the use of XML Signature and XML Encryption, WSS aims to offer client/service security.

- **WS-SecureConversion** is a Web services protocol for creating and sharing security context. WS-SecureConversion is meant to operate in systems where WS-Security, WS-Trust, and WS-Policy are in use, and it attaches a security context token to communications such as SOAP used to transport messages in an SOA enterprise.

- **WS-SecurityPolicy** provides a set of network policies that extend WS-Security, WS-Trust, and WS-SecureConversion so messages complying to a policy must be signed and encrypted. The SecurityPolicy is part of a general WS-Policy framework.

- **WS-Trust extends WS-Security** to provide a mechanism to issue, renew, and validate security tokens.

- A Web service using WS-Trust can implement this system through the use of a Security Token Service (STS), a mechanism for attaching security tokens to messages and a set of mechanisms for key exchanges that are used to validate tokens and messages.

- Another approach to enforcing security in SOA is to use an XML gateway that intercepts XML messages transported by SOAP or REST, identifies the source of the message, and verifies that the message was securely received.

- Providing XML Gateway SOA security requires a Public Key Infrastructure (PKI) so that encryption is enforced by digital signatures.

- Progress Software's Actional 8.0 (http://web.progress.com/en/actional/index.html) now includes Mindreef's SOAPscope Server and has an XML middleware service that performs diagnostic testing and Web services governance, adding that component to Actional's ability to monitor and map XML appliances and application servers.

The Open Cloud Consortium

- The Open Cloud Consortium (OCC; see http://opencloudconsortium.org/) is an organization comprised of several universities and interested companies that supports the development of standards for cloud computing and for interoperating with the various frameworks.

OCC working groups perform these functions:

- They develop benchmarks for measuring cloud computing performance. Their benchmark and data generator for measuring large data clouds is called MalStone (http://code.google.com/p/malgen/).
- They provide testbeds that vendors can use to test their applications, including the Open Cloud Testbed and the Intercloud Testbed that are part of the work of the Open Cloud Testbed and Intercloud working groups.
- They support the development of open-source reference implementations for cloud computing.
- The Working Group on Standards and Interoperability For Large Data Clouds extends the architecture for data storage with a distributed file system, table services, and computing using MapReduce following the model that is part of Google's offering.

The Open Cloud Consortium

- MapReduce is Google's patented software framework that supports distributed large data sets organized by the Google File System (GFS) accessed by clusters of computers.

- The Apache Hadoop (http://hadoop.apache.org/) open-source system is based on MapReduce and GFS.

- They support the management of cloud computing infrastructure for scientific research as part of the Open Science Data Cloud (OSDCP) Working Group's initiative.

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- SOA components are often best-of-breed service providers that can provide a measured service level and can play a role in Business Process Management (BPM) systems.
- The separation of services from their design allows for much easier system upgrades and maintenance. Many Web 2.0 applications use SOA components, and SOA will become increasingly useful in larger applications that require many Web services.
- These applications often rely on REST and feature AJAX components in a user interface that supports Web syndication blogs, and wikis. Some people regard mashups as Web 2.0 applications as well.
- A mashup is the combination of data from two or more sources that creates a unique service. The layers added to Google maps are examples of mashups.
- AJAX stands for Asynchronous JavaScipt and XM. AJAX is a set of development tools that allow for client input into Web applications; it is not a standard.
- AJAX describes a group of technologies that leverage HTML and CSS for styling, Web objects in the Document Object Model (DOM) or data, XML and XSLT for data interchange, the XMLHttpRequest for asynchronous communication, and JavaScript commands to request data from data sources.

Relating SOA and Cloud Computing

- The challenge SOA faces in designing systems to support Web 2.0 is the lack of standardization in how components in Web 2.0 are used.
- People believe that SOA will play a role in creating what has been dubbed an “Internet of Services” where complex services will be available for use as a set of building blocks based on the convergence of SOA and Web 2.0.
- The Gartner Group refers to this trend as the development of “Advanced SOA,” but features of SOA that are event-driven have been part of many vendors’ middleware offerings for several years now.